

Transfer Learning

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Why Train From Scratch Is Expensive

The data-hungry, compute-intensive costs from earlier, made concrete



TRAINING FROM SCRATCH

A model like ResNet-50 was trained on roughly 1.2 million labeled images, using days of multi-GPU compute — before it could recognize anything useful.



YOUR ACTUAL PROBLEM

A few hundred labeled weld-defect images, and a training budget measured in hours on a single GPU, not days on a cluster.

That gap is exactly what transfer learning is designed to close.

The Core Idea: Reuse, Don't Restart

Start from a model that already learned general-purpose patterns

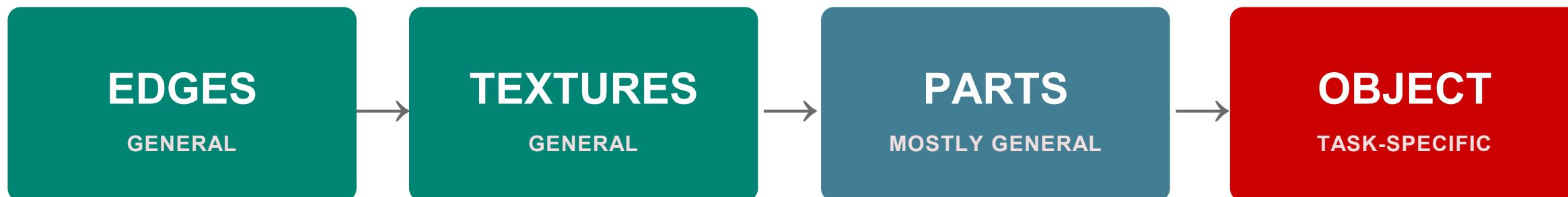


“Pre-trained” just means someone already paid the data-and-compute cost — you’re borrowing the result.

Why It Works: Not All Layers Are Equally Task-Specific

A direct callback to the depth-and-hierarchy idea from the CNN session

Reusable across almost any task → → → Needs adaptation for your specific task



This is why you can freeze the early layers and only retrain the last one or two — they were never task-specific to begin with.

Two Strategies: Feature Extraction vs. Fine-Tuning

Different amounts of the pretrained model get updated



Feature Extraction

Freeze the pretrained layers entirely; only train a small new “head” on top for your specific task.

Best when: your dataset is small and similar to the pretraining data.



Fine-Tuning

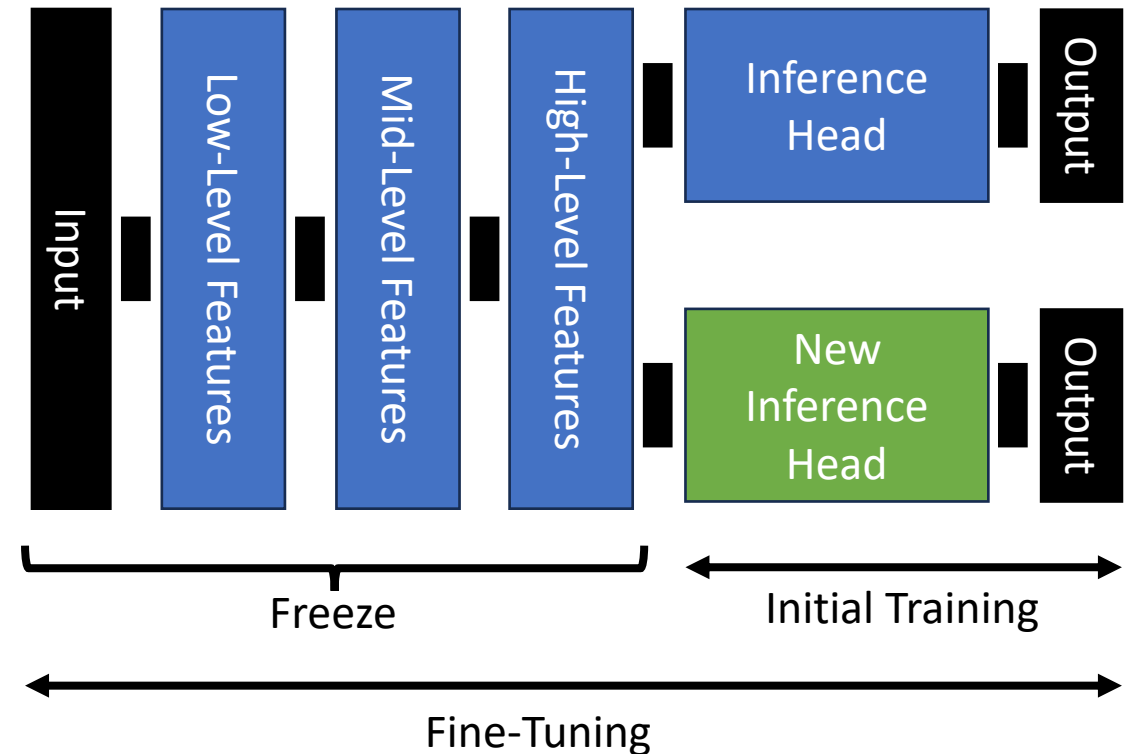
Unfreeze some or all pretrained layers and keep training them, usually at a lower learning rate.

Best when: you have more labeled data, or your task differs more from the pretraining data.

Both start from the same pretrained checkpoint — the only difference is how much of it you let training touch.

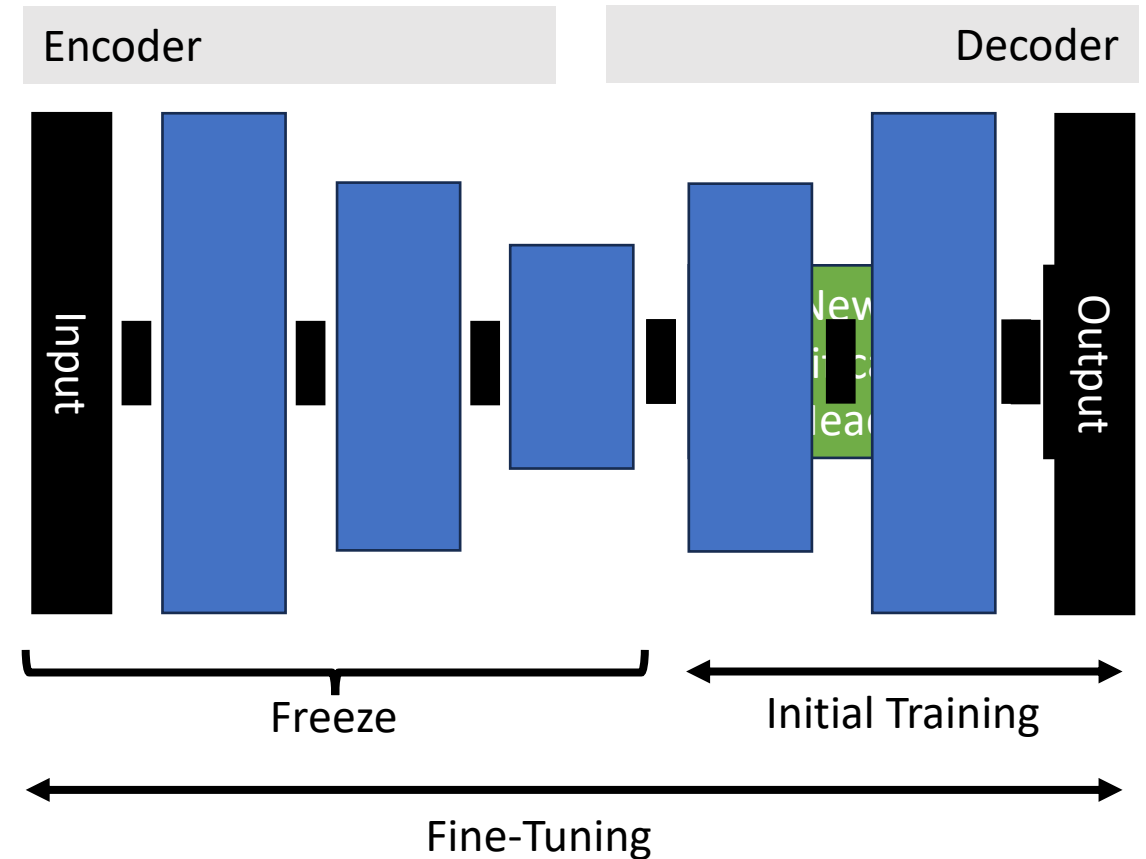
Transfer Learning: Across Datasets

- Across datasets of the same format, we can **pre-trained** model by removing its head and replacing it with a new head.
- The new head is trained by **freezing** the rest of the network.
- This is often followed by **fine-tuning** of the entire network.



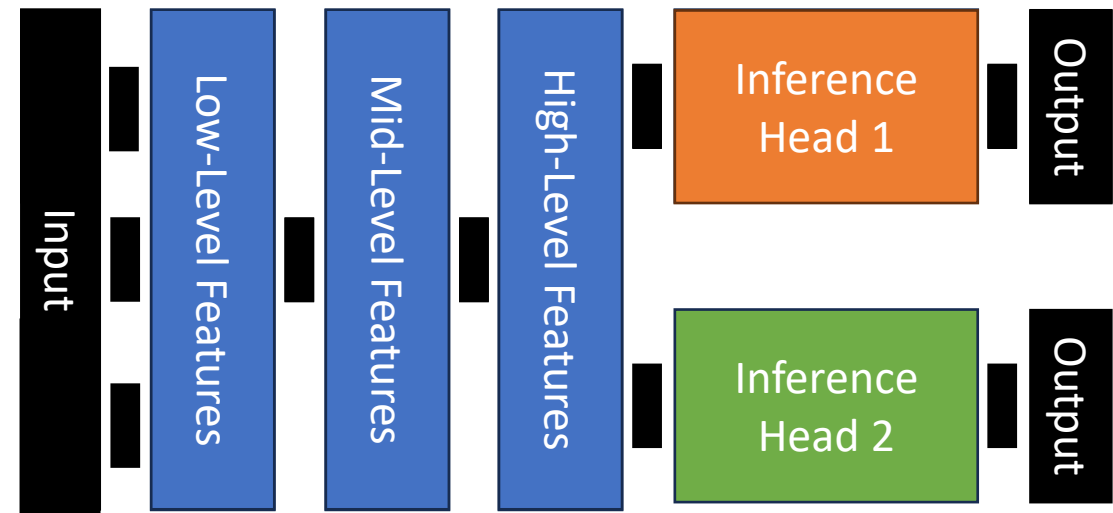
Transfer Learning: Semi Supervised

- Autoencoders are an option when there is no labeled data available in the same format from other datasets, but there is enough unlabeled data.
- We use the encoder portion of the network and attach an inference head.



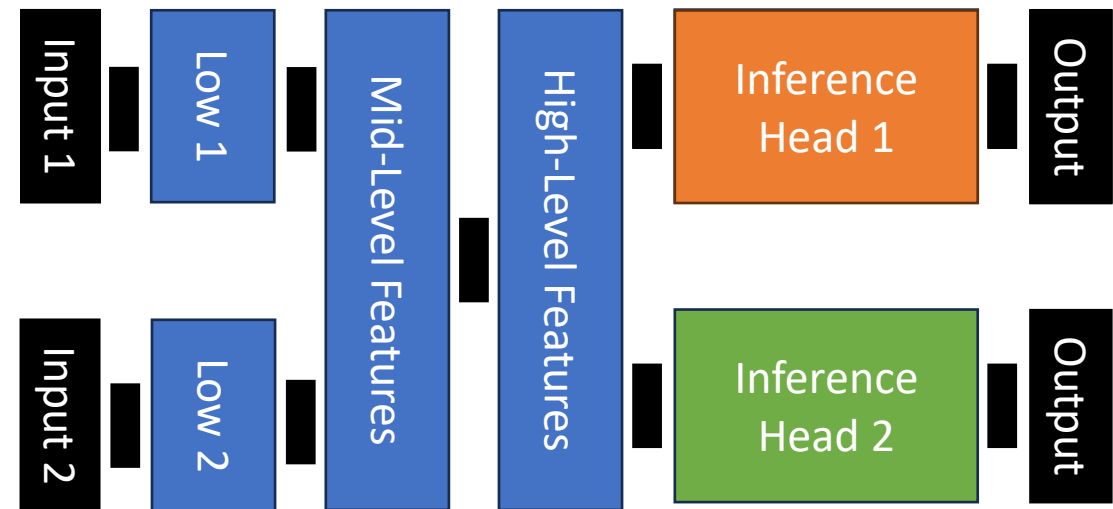
Multi-Task Learning

- It can be used to train multiple inference tasks jointly so they can help each other.
- Useful when one or all the datasets are relatively small.
- The cost is the sum of the cost for each inference task separately applied to their corresponding dataset. During training we use a mixed batch.



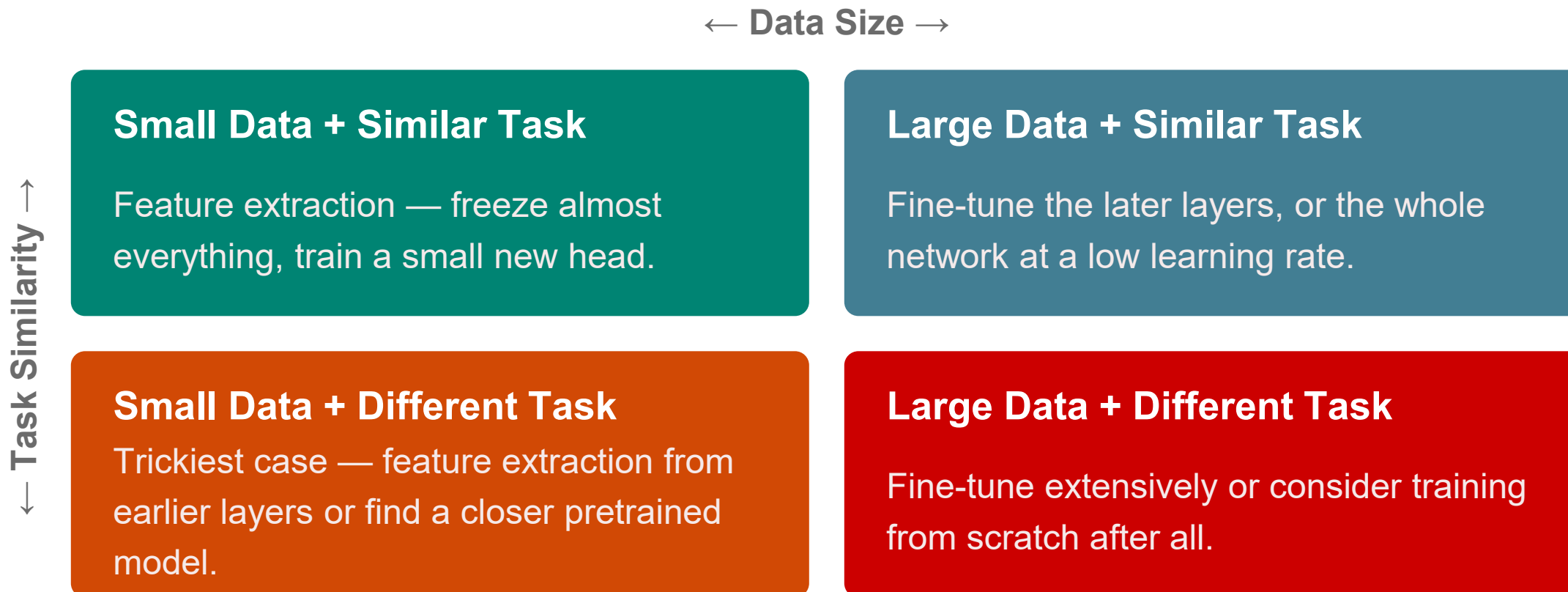
Multi-Task Learning

- Some variants may incorporate separate learning of low-level features.
- Useful when considering images of different modalities for similar tasks, e.g., using infrared images for detection of people, and color images for general object detection.



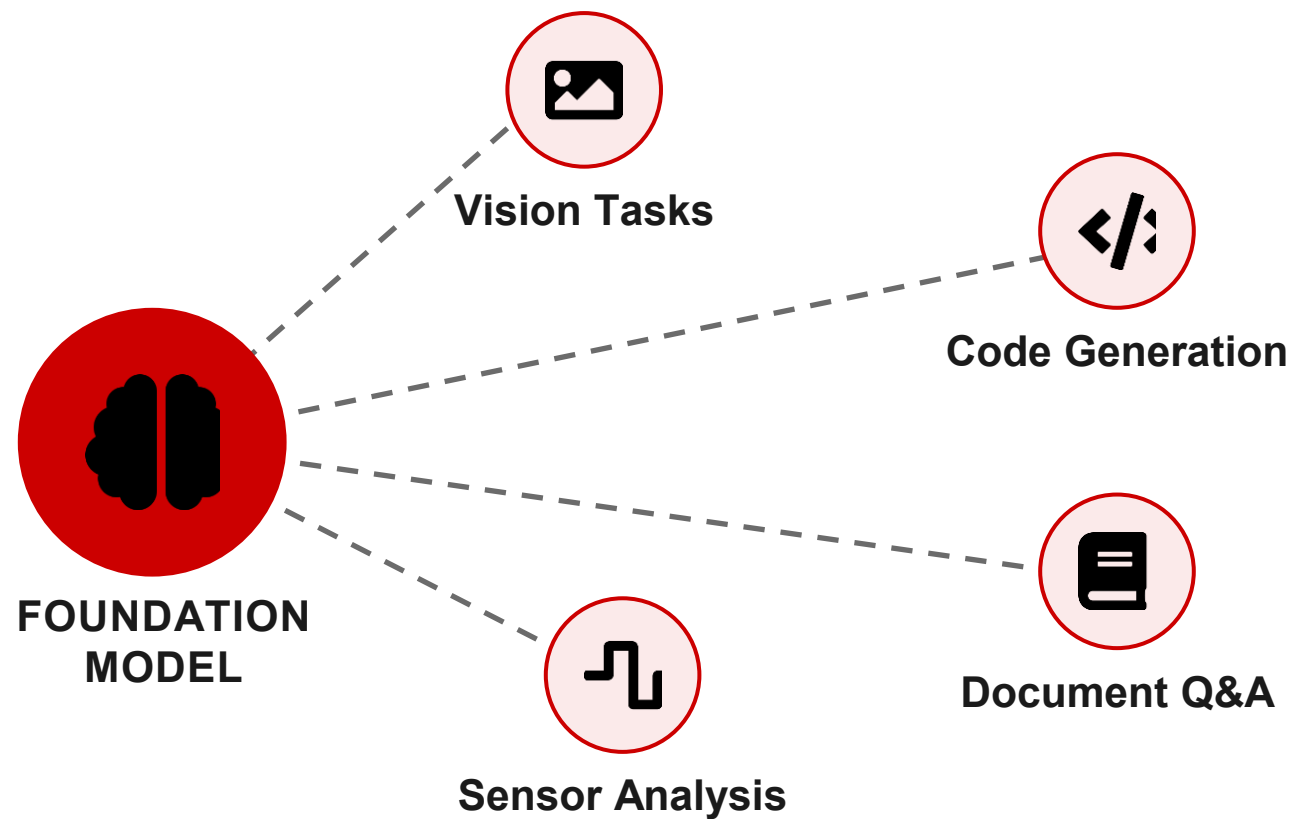
Choosing a Strategy: Data × Task Similarity

A simple decision framework for which approach to reach for



Foundation Models: The Current Extreme

One massive pretrained model, adapted across countless downstream tasks



Same underlying idea as before, taken to the extreme: one very expensive pretraining run, reused millions of times over.

Limitations & Watch-Outs

Transfer learning isn't automatic — and it isn't free



Domain Mismatch

A model pretrained on natural photos may not transfer well to thermal or X-ray imagery — the features it learned may not apply.



Catastrophic Forgetting

Fine-tuning too aggressively, or on too little data, can erase the general knowledge you were trying to reuse.



Licensing & Compute

Large foundation models still carry real compute and licensing costs for fine-tuning and deployment.